Keeper tier-gate — Grace Periodic Table v0.7 (stability-class 6th axis absorbing Lyra #397) + Lyra Strong-Uniqueness Theorem v1.1 (C18 measure-as-theorem + C19 ρ-pinning + C2 corrected + Route-A). Periodic Table v0.7: PASS at FRAMEWORK; absorbs Quasi-Eigentone N1 cleanly via mass-eigenstate-vs-flavor framing. Strong-Uniqueness v1.1: PASS at STRENGTHENED tier; null-model recompute filed per Lyra Section 6.1 request — single-independent-residual framing, NOT multiplicative criterion count.

Keeper

2026-05-30 Saturday morning batch follow-up

Tier-gate — Periodic Table v0.7 + Strong-Uniqueness v1.1

1. Grace Periodic Table v0.7

Verdict: PASS at FRAMEWORK. Clean v0.6 → v0.7 incremental absorption of Lyra Quasi-Eigentone Framework v0.1 (#397) as the 6th per-cell axis (Stability class).

Strengths: - Per-cell tagging consistent: fundamental block (4 cells) + lepton row (with mass-eigenstate-vs-flavor caveat) + gauge sector + composite block (6 cells from v0.6). - Mass-eigenstate-vs-flavor-basis framing for neutrinos resolves my prior N1 to Lyra cleanly: "Note on neutrinos: oscillations complicate the 'stable' assignment in the flavor basis. In the mass-eigenstate basis, the lightest neutrino is TRUE; the heavier ones oscillate but don't decay (no charge-conserving channel to electron-flavored leptons + photon)." This is exactly the right framing — Grace has independently arrived at the granularity I recommended. - Five-Absence re-framed as structural "no new K-type bottoms beyond established" — clean reformulation. - Proton stability gets explicit substrate-derivation: "TRUE EIGENTONE = lowest baryon in V_(1,1) bulk k=6 closure; no grading-allowed decay channel exists; quark content cannot reorganize into lower-mass color-singlet by Green coproduct + Q/B/L conservation." This is the substrate-derived proton stability claim made explicit. - Predictions from v0.7 (Section G): proton lifetime infinite; gluon never in isolation; no two degenerate true-eigentones at same K-type; Casimir-twin V_(3,3)+V_(4,1) both QUASI (no-SUSY-without-SUSY signature); tensor glueball at Casimir 16 QUASI. Clean, falsifiable. - Standing conventions (5) maintained from v0.6. - Composite block stability tags appropriate (all QUASI for hadron resonances; mixed long-lived for J/ψ/Y noted honestly).

Notes (none load-bearing — small refinements):

- Section C gauge sector entry: "photon γ is V_(1,0), not V_(1,1); included here for SM-gauge-sector coverage." This is the right disambiguation already handled.
- Section E speculative cells 12-18 — appropriately tagged speculative.

Tier disposition: - 6-tuple per-cell taxonomy with stability class added: FRAMEWORK (consistent extension)

- Per-cell stability tagging (fundamental + lepton + gauge + composite blocks): FRAMEWORK with sub-tier caveats as Grace marked - Mass-eigenstate-vs-flavor ν resolution: FRAMEWORK — addresses prior Keeper N1 - Section G predictions: FRAMEWORK — sharp + falsifiable

Status: PASS at FRAMEWORK; no new conditions; absorbs my prior N1 to Quasi-Eigentone cleanly. v0.7 is the right incremental step.

2. Lyra Strong-Uniqueness Theorem v1.1

Verdict: PASS at STRENGTHENED tier. C18 + C19 + C2 correction + Route-A + (3,3,5) forcing chain advance the theorem substantively. Critically, Lyra's honest correction in Section 6 (uniqueness strengthens by REDUCING independent choices, not adding criteria) is the right referee-quality framing — caught by Keeper before referees, corrected before external use.

Strengths:

- C18 (FK-measure FORCED by Born/Gleason) is a derived theorem strengthening the uniqueness claim: removes the measure as a free input rather than adding a separate criterion. The framing "this is a uniqueness criterion of a new KIND" is correct: not "D_IV^5 uniquely satisfies property X" but "given D_IV^5, the quantum structure is forced."
- C19 (ρ-vector pins three primaries): ρ = (n_C, N_c)/rank = (5/2, 3/2); rank, n_C, N_c reduce to one canonical invariant (ρ) + region structure (bulk ρ_1 = n_C/rank; Shilov ρ_2 = N_c/rank). This is rigorous B_2 representation theory for the value; the "pins three primaries to one invariant" framing is the FRAMEWORK-PLUS interpretation as Lyra correctly tagged.
- C2 corrected to 5/2 = ρ_1 (was 7/2 mislabel) is verified via Elie Toy 3582 + literature. Correction is substantively load-bearing for ρ-pinning argument.
- Route-A "choose D_IV^5, zero inputs" is the strongest form of the uniqueness claim: five integers ARE invariants of one domain, not independent choices.
- (3,3,5) → D_IV^5 forcing chain (Section 6.2, Elie Toys 3589-3591): independent uniqueness leg from SM bare data. Lyra correctly marks as CONDITIONAL on B1 closure (generation-mechanism multi-week burden).
- Section 6 honest correction: "My first v1.1 pass implied 'more criteria → smaller null model.' That is a coincidence-inflation error — Keeper held the line." This is the right discipline — Cal #27 + Calibration #34 (headline-cap-conditionality) firing internally before external use. Excellent.
- v1.2 patch absorbs Cal #156 Flags A+B + Cal #99: C13 mis-attributed to T2468 corrected (SCMP is T2469); T2467 Rigidity-as-Singleton is META-tier (uniqueness frame), not operational criterion. Operational count corrected from 14 → 13 criteria + 1 META.
- Second N_max representation (Elie Toy 3594): N_max = 2^g + N_c^2 = 128 + 9 = 137. Second exact form complements the definitional $N\_c^3 \cdot n\_C$ + rank = 137; ties N_max to 2^g (Hall-Littlewood field-size structure).

Null-model recompute (Keeper-graded, per Lyra Section 6.1 request):

The corrected framing: count only INDEPENDENT structural choices, not derived consequences. Per Elie Toy 3592 dependency DAG, the independent residual is:

Layer	Independent?	Reason
Type-IV bounded symmetric domain at complex dim n_C = 5	INDEPENDENT (the single primary choice)	The domain choice itself
rank = 2	DERIVED from type-IV at complex dim 5	Type-IV ranks are rank $\leq$ 2; complex dim 5 forces rank 2
N_c = 3 (h^∨)	DERIVED from rank=2 + SO(5)=B_2	h^∨(B_2) = 3
C_2 = 6 (Casimir)	DERIVED from B_2	B_2 Casimir
g = 7 (embedding)	DERIVED from type-IV at dim 5	p+q = 5+2 = 7
N_max = 137	DERIVED (definitional)	$N\_c^3 \cdot n\_C$ + rank

Layer	Independent?	Reason
c_FK · π^(9/2) = 225	DERIVED (Born/Gleason theorem, C18)	Measure forced
ρ_1 = 5/2, ρ_2 = 3/2	DERIVED (B_2 Weyl vector / rank, C19)	ρ-pinning

Net independent residual: ONE choice — "the domain is type-IV bounded symmetric domain at complex dim n_C = 5."

The honest null-model factor is thus not (1/3)^14; it is the probability that a random bounded symmetric domain matches this single choice + that the matching domain's invariants reproduce SM gauge structure.

Order-of-magnitude estimate: - Bounded symmetric domains by type: I, II, III, IV (classical) + V, VI (exceptional). Six families. - Within type IV: indexed by complex dimension n ≥ 2. - Selecting a single type-IV at single dim = ~1/N where N is the count of "plausible" candidates SM-compatible (rank-2 + dim-5 substantively narrows to D_IV^5). - Combined with the (3,3,5) → D_IV^5 forcing chain (Section 6.2): SM-known (3 colors, 3 generations) ⇒ rank-2 + h^∨=3, of which B_2 is one of {A_2, B_2≅C_2, G_2} = effectively 3 rank-2 candidates. dim 5 then picks B_2 within those that admit dim-5 representations cleanly.

Honest null-model order-of-magnitude: ~1/10 to ~1/100 — single dominant factor, NOT (1/3)^14 = 2×10^{-7}.

The strength of the uniqueness claim is in the CONVERGENCE OF INDEPENDENT ROUTES, not in multiplicative criterion count. v1.1 documents multiple independent routes (ρ-pinning + Route-A + (3,3,5) forcing + cyclotomic/Mersenne network + Hall-Littlewood-corner) converging on D_IV^5. This convergence is the strongest evidence — not a single number on a null model.

Recommendation for v1.2: state the null model in the corrected single-residual framing explicitly; do NOT cite (1/3)^N criterion-count framings; emphasize convergence-of-routes as the structural evidence.

Tier disposition: - C1-C13 RIGOROUSLY CLOSED (per v1.2 patch operational count): CARRIED - META T2467 Rigidity-as-Singleton: META-tier (uniqueness frame, not operational criterion) - C2 corrected value 5/2 = ρ_1: RIGOROUSLY CLOSED - C18 (FK-measure FORCED): RIGOROUSLY CLOSED as derived theorem - C19 (ρ-vector pinning): FRAMEWORK-PLUS as Lyra correctly tagged - Route-A reduction: STRUCTURAL STRENGTHENING acknowledged - (3,3,5) → D_IV^5 forcing chain: CONDITIONAL PASS gating on B1 (generation-mechanism) closure - Null-model framing (corrected): PASS under single-residual framing; recommend v1.2 doc update explicit - C15-C17 candidates: OPEN (multi-week / Casey-decision)

Status: PASS at STRENGTHENED tier; null-model recompute filed; v1.2 doc-update recommended for explicit single-residual framing + convergence-of-routes emphasis.

3. Routed handback

- Lyra: Strong-Uniqueness v1.2 absorbs (a) Keeper null-model recompute in Section 6 with explicit single-residual framing + convergence-of-routes emphasis; (b) Cal cold-read on C18 + C19 (load-bearing new theorems). Periodic Table v0.7 mass-eigenstate-vs-flavor ν framing already resolves my Quasi-Eigentone v0.1 N1; can be reflected in Quasi-Eigentone v0.2.
- Grace: Periodic Table v0.7 PASS; no new conditions. v0.8 follows when bulk-color closes (per-cell quark/gauge stability resolves via gauge-phenomenology emergence).
- Cal: cold-read queue extends — Strong-Uniqueness v1.1 C18 (Born/Gleason measure theorem) + C19 (ρ-vector pinning interpretation) + null-model recompute. v1.2 with corrected framing pending Lyra.
- Elie: Toy 3592 dependency DAG + Toy 3594 second-N_max identity are the load-bearing computational anchors for v1.1; both stand. When engine v0.3 V-compound layer lands, Vol 16 §3 v0.1 + Periodic Table v0.7 composite block stability tags get RIGOROUS upgrade from FRAMEWORK.

4. Queue continuation

Saturday batch tier-gates now: 9 items dispositioned (7 afternoon + 2 morning re-verification: Grace v0.7 + Lyra Strong-Uniqueness v1.1).

Remaining in immediate Saturday morning queue (not yet tier-gated): - Grace G12 Hadron Composite K-Type Mendeleev v0.2 - Grace G13 8-Gluon Emergence Catalog Scaffold v0.1 - Grace Substrate-Primary Arithmetic Mendeleev v0.1 (precursor to Two-Route Scan) - Grace 4.5 K-Types-Are-Particles Falsifier Design v0.1 - Grace 4.1 SO(5) Shell-Closure Catalog Scaffold v0.1 - Grace 1.5 Two-Structures Mendeleev Pair Analysis v0.1 - Lyra C12/C14 T-Number META Audit Patch v0.1 (referenced in v1.1) - Lyra K-Types-Are-Particles Falsifier Design v0.1 - Lyra P43 Holomorphic Discrete Series K-Type Framework v0.1 - Cal Proposal K-Types-Are-Particles Falsifier Design v0.1 (cross-CI triad with Lyra + Grace) - Lyra A1 v0.4 Final Disposition v0.1 - Lyra Saturday Afternoon Consolidation - Lyra E9 Dictionary Read v0.1

Pulling next: the K-types-are-particles falsifier triad (Cal proposal + Lyra design + Grace design) — three CIs converging on the same falsifier suggests load-bearing structural cross-check; tier-gate all three together.

— Keeper. Two more tier-gates (Periodic Table v0.7 + Strong-Uniqueness v1.1); null-model recompute delivered per Lyra Section 6.1 request; uniqueness framing corrected to convergence-of-routes; queue advances to K-types-are-particles falsifier triad next.